

Moni Kumari

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PROFESSIONAL SUMMARY

Computer Science Engineering student specializing in Data Science with strong foundations in **C++**, **OOP**, and **DSA**, and hands-on **software engineering** experience building production software. **Built and launched a SaaS mobile app with integrated payments and paying subscribers**, handling billing, reconciliation, and webhook verification. Built production React Native and Next.js applications to a from-scratch SQL query engine achieving **5× faster performance than SQLite**.

TECHNICAL SKILLS

- **Languages:** C++, Python, TypeScript, JavaScript, SQL, HTML5, CSS3
- **Core CS:** Object-Oriented Design, DSA, Complexity Analysis, System Design, Query Optimization
- **Frontend Development:** React, Next.js, React Native (Expo), Tailwind CSS, Responsive UI
- **Backend Development:** Node.js, Express.js, REST APIs, Cloud Functions, JWT Auth, Webhooks, Payment Integration
- **Databases:** Firebase, Supabase, MongoDB, SQLite
- **DevOps & Tooling:** Git, GitHub Actions, Docker, Vercel, EAS/Expo, Postman, Deployment Workflows

EDUCATION

CMR Institute of Technology

Bachelor of Engineering – Computer Science & Engineering (Data Science), CGPA: 8.86

Bengaluru, India

2024 – 2028

PROJECTS

UniTrack – Mobile Subscription & Payments SaaS

[Live](#) | [GitHub](#)

- **Stack:** React Native (Expo), TypeScript, Firebase, Razorpay, Gemini API
- Launched to **20+ active users with paying subscribers**, generating recurring revenue.
- Built Razorpay billing across 3 tiers with verified webhooks, payment reconciliation and failed-transaction recovery.
- Shipped an admin dashboard for revenue analytics and subscription health on Firebase Cloud Functions.
- Cut onboarding from **15 minutes to 30 seconds** using Gemini vision to extract timetables from photos.

Quarry – Columnar Analytics Query Engine

[GitHub](#)

- **Stack:** Python, NumPy, SQL, Object-Oriented Design, Systems Programming
- Built a SQL engine from scratch — lexer, parser, planner, vectorized executor — **5x faster than SQLite** on 1M rows.
- Debugged and profiled a GROUP BY performance bottleneck, then added dictionary encoding: **731ms to 53ms, a 13.8x improvement**.
- Separated storage, planning and execution into clean layers, allowing projection pushdown without parser changes.
- Validated by **differential testing against SQLite**, catching 2 semantic bugs; 68 tests passing.

goIRL – Tech Events Discovery Platform

[Live](#) | [GitHub](#)

- **Stack:** Next.js 16, TypeScript, Supabase (PostgreSQL), MapLibre GL, Vercel
- Aggregated events from the MLH, Lu.ma and Devfolio APIs into one searchable platform with a 3D map.
- Implemented magic-link auth, PostgreSQL full-text search, an admin moderation panel and .ics export.
- Automated daily ingestion across 3 external sources using Vercel cron jobs.

ACHIEVEMENTS & LEADERSHIP

- **Open Source Contributor** — The-Boring-Education/TBE-Web: shipped a pull request to production through main-tainer **code review** in a TypeScript/Next.js monorepo serving active users.
- **INNOVATEX 4.0 Hackathon** | Presidency University: Built **2 working prototypes in 9 hours** (2025)

CERTIFICATIONS

- **C++ Data Structures & Algorithms** + **LeetCode** (Udemy, 2025)
- **Python Hands-On, 210 exercises** (Udemy, 2025)